



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

DEPARTMENT OF ROBOTICS AND AI

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OOP LAB

Project Report

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Submitted to	Hasaan Mujtaba
Name	Registration number
Abdul Samad	25108225
Hussain Khan	25108244
Syed Shaheer Ali Shah	25108276

• **Digital Clinic Prescription and Health Tracker**

Highlights BMI Features

Introduction

Object Oriented Programming (OOP) is a modern programming approach that helps in organizing code into classes and objects. It makes programs more structured, reusable, and easy to manage. In this project, we have developed a Clinic Prescription System using the C++ programming language. This system simulates a real-world clinic environment where patient details are recorded and prescriptions are generated automatically. The program allows the user to enter patient information such as name, height, weight, symptoms, and medicine. It then calculates the Body Mass Index (BMI), displays a formatted prescription, and stores the data in a file for future use.

Purpose of Project

The main purpose of this project is to demonstrate how Object-Oriented Programming concepts can be applied to solve real life problems.

This system is useful for:

- Managing patient records
- Generating prescriptions
- Reducing manual paperwork
- Organizing medical data efficiently
- It provides a simple and effective way to handle clinic related tasks using programming.

Objectives of Project

The objectives of this project are:

- To understand and implement OOP concepts in C++
- To design a real-life based system
- To store and manage patient information
- To calculate BMI and analyse patient health
- To save records permanently using file handling
- To create a user friendly and structured program

Overview of System

The Clinic Prescription System works in the following way:

- The program starts and displays a welcome message
- The user enters the number of patients
- For each patient, details are entered
- The system calculates BMI automatically
- The prescription is displayed in a formatted way
- The data is saved into a file named clinic.txt
- The process repeats for multiple patients
- This system helps simulate how clinics manage patient prescriptions digitally.

Features of System

This project includes the following features:

- **Patient Information Entry**
 1. Name, Height, Weight, Fever condition, Symptoms
- **Medicine Details**
 2. Medicine name
 3. Usage (morning, afternoon, night)
 4. Duration (e.g., 1 week, 5 days)
- **Automatic Functions**
 1. BMI calculation
 2. Health status (Normal, Underweight, etc.)
 3. Current date and day
 4. Doctor name automatically added
 5. Data Management
 6. Displays formatted prescription
 7. Saves data into file
 8. Supports multiple patients

OOP Concepts Used

- **Encapsulation**

Encapsulation means keeping data private and accessing it through functions. In this project, all patient data is declared as private, and functions like `input()` and `display()` are used to access it.

- **Class and Object**

A class named Patient is created. It acts as a blueprint. An object of this class is used to perform operations like taking input and displaying data.

- **Abstraction**

Only important details are shown to the user, while complex calculations like BMI are hidden inside functions

- **Modularity**

The program is divided into small functions like:

1. input()
2. display()
3. calculateBMI()
4. saveToFile()

This makes the program easy to understand and manage.

BMI Calculation

It uses age-based standard weight ranges for better health analysis:

- 1–5 years: 12–20 kg
- 6–10 years: 20–32 kg
- 11–15 years: 32–55 kg
- 16–20 years: 50–75 kg
- 21+ years: Standard BMI formula

This makes health analysis more accurate according to age.

The system categorizes BMI into:

- Underweight
- Normal
- Overweight
- Obese

This helps in analysing the patient's health condition.

File Handling

- File handling is used to store data permanently.
- The program uses ofstream
- Data is stored in clinic.txt
- ios:app mode is used to append data
- This ensures that previous records are not deleted and all patient data is saved

Working of Program

- Program starts
- User inputs number of patients
- Loop runs for each patient
- Data is entered
- BMI is calculated
- Prescription is displayed
- Data is saved in file
- Program ends

Advantages of System

- Easy to use
- Saves time
- Reduces manual work
- Stores patient data permanently
- Provides health analysis (BMI)
- Organized and professional output

Limitations of System

- No graphical user interface (GUI)
- Cannot edit or delete records
- Basic functionality only
- Requires manual input

Applications of System

This system can be used in:

- Small clinics
- School medical rooms
- Training and educational purposes
- Basic hospital management systems

Future Improvements

This system can be improved by adding:

- Graphical User Interface (GUI)
- Database integration
- Search patient record option
- Delete/update functionality
- Login system for doctors
- Online prescription system

Comparison with Traditional Paper Prescription

In traditional clinics, prescriptions are usually written on paper. While this method has been used for many years, it has several limitations. The computerized Clinic Prescription System provides many improvements over the manual paper-based method.

- **Accuracy**
 1. Paper System: Handwriting can be unclear and may cause mistakes.
 2. Our System: Data is typed clearly, reducing errors and confusion.
- **Data Storage**
 1. Paper System: Prescriptions can be lost, damaged, or misplaced.
 2. Our System: Data is stored in a file (clinic.txt) and remains safe.
- **Speed**
 1. Paper System: Writing everything manually takes more time.
 2. Our System: Faster input and automatic formatting save time.
 3. Organization

- a. Paper System: Hard to manage many patients records
- b. . Our System: All records are stored in an organized digital format.

- **Additional Features**

1. Paper System: No automatic calculations.
2. Our System: Automatically calculates BMI and shows health status.

- **Professional Look**

1. Paper System: Depends on handwriting and format.
2. Our System: Clean, structured, and professional prescription format.

Code

```
#include <iostream>
#include <fstream>
#include <iomanip>
using namespace std;

class Person {
protected:
    string name;
    int age;

public:
    void inputPerson() {
        cin.ignore();
        cout << "Enter Patient Name: ";
        getline(cin, name);

        cout << "Enter Age: ";
        cin >> age;
    }
};
```

```
class HealthTracker : public Person {
private:
    float height, weight, fever;
    string condition, symptoms;
    string medicineName, duration;
    string morning, afternoon, night;
    string allergies;

    string doctorName = "Dr. Ahmed Khan";
    string hospitalName = "SHS CLINIC";

    string checkDay, checkDate, checkTime;

public:

    void inputDetails() {
        inputPerson();
        cin.ignore();

        cout << "Enter Patient Checking Day : ";
        getline(cin, checkDay);

        cout << "Enter Patient Checking Date : ";
        getline(cin, checkDate);

        cout << "Enter Patient Checking Time : ";
        getline(cin, checkTime);

        cout << "Enter Height (meters): ";
        cin >> height;

        cout << "Enter Weight (kg): ";
        cin >> weight;

        cout << "Enter Fever (celcius): ";
        cin >> fever;
```

```
cin.ignore();

cout << "Enter Condition: ";
getline(cin, condition);

cout << "Enter Symptoms: ";
getline(cin, symptoms);

cout << "Enter Medicine Name: ";
getline(cin, medicineName);

cout << "Morning Dose: ";
getline(cin, morning);

cout << "Afternoon Dose: ";
getline(cin, afternoon);

cout << "Night Dose: ";
getline(cin, night);

cout << "Duration: ";
getline(cin, duration);

cout << "Allergies: ";
getline(cin, allergies);
}

float calculateBMI() {
    return weight / (height * height);
}

string bmiStatus() {
    float bmi = calculateBMI();

    if(age >=1 && age<=5){
        if(weight<12) return "Underweight";
        else if(weight<=20) return "Normal Weight";
```

```

        else return "Overweight";
    }

    else if(age>=6 && age<=10){
        if(weight<20) return "Underweight";
        else if(weight<=32) return "Normal Weight";
        else return "Overweight";
    }

    else if(age>=11 && age<=15){
        if(weight<32) return "Underweight";
        else if(weight<=55) return "Normal Weight";
        else return "Overweight";
    }

    else if(age>=16 && age<=20){

        if(weight<50) return "Underweight";
        else if(weight<=75) return "Normal Weight";
        else return "Overweight";
    }

    else{
        if(bmi<18.5) return "Underweight";
        else if(bmi<=24.9) return "Normal Weight";
        else if(bmi<=29.9) return "Overweight";
        else return "Obese";
    }
}

void displayPrescription() {

    cout << endl;
    cout << hospitalName << endl;
    cout << "DIGITAL PRESCRIPTION" << endl;
    cout << endl;

```

```
cout << "Doctor Name      : " << doctorName << endl;
cout << "Patient Checking Day : " << checkDay << endl;
cout << "Patient Checking Date : " << checkDate << endl;
cout << "Patient Checking Time : " << checkTime << endl;
```

```
cout << endl;
```

```
cout << "Patient Name      : " << name << endl;
cout << "Patient Age       : " << age << endl;
cout << "Patient Height    : " << height << " m" << endl;
cout << "Patient Weight     : " << weight << " kg" << endl;
cout << "Patient Fever      : " << fever << " celcius" << endl;
cout << "Patient Condition  : " << condition << endl;
cout << "Patient Symptoms    : " << symptoms << endl;
```

```
cout << "Patient BMI      : "
    << fixed << setprecision(2)
```

```
    << calculateBMI() << endl;
```

```
cout << "Health Status    : "
    << bmiStatus() << endl;
```

```
cout << endl;
```

```
cout << "Medicine Name      : " << medicineName << endl;
cout << "Morning Dose       : " << morning << endl;
cout << "Afternoon Dose      : " << afternoon << endl;
cout << "Night Dose          : " << night << endl;
cout << "Duration            : " << duration << endl;
cout << "Allergies           : " << allergies << endl;
```

```
cout << endl;
```

```
}
```

```
void saveToFile() {
    ofstream file("clinic.txt", ios::app);
```

```

        file << hospitalName << endl;
        file << "Doctor Name: " << doctorName << endl;
        file << "Patient Checking Day: " << checkDay << endl;
        file << "Patient Checking Date: " << checkDate << endl;
        file << "Patient Checking Time: " << checkTime << endl;
        file << "Patient Name: " << name << endl;
        file << "Age: " << age << endl;
        file << "Height: " << height << endl;
        file << "Weight: " << weight << endl;
        file << "BMI: " << calculateBMI() << endl;
        file << "Health Status: " << bmiStatus() << endl;
        file << "Medicine: " << medicineName << endl;
        file << "Allergies: " << allergies << endl;
        file << endl;

        file.close();
    }

};

int main() {

    int patients;

    cout << "WELCOME TO SHS CLINIC SYSTEM" << endl;
    cout << "Enter Number of Patients: ";
    cin >> patients;

    for(int i=1;i<=patients;i++) {

        HealthTracker p;

        cout << endl;
        cout << "Patient " << i << endl;

        p.inputDetails();
        p.displayPrescription();
    }
}

```

```
        p.saveToFile();  
    }  
  
    cout << endl;  
    cout << "All Records Saved Successfully!" << endl;  
  
    return 0;  
}
```

Output:

WELCOME TO SHS CLINIC SYSTEM

Enter Number of Patients: 097645674

Patient 1

Enter Patient Name: saim ayyub

Enter Age: 18

Enter Patient Checking Day : monday

Enter Patient Checking Date : 04-03-26

Enter Patient Checking Time : 2PM

Enter Height (meters): 1.83

Enter Weight (kg): 65

Enter Fever (celcius): 101

Enter Condition: high fever

Enter Symptoms: cough and throat infection

Enter Medicine Name: burifen tab and panadol

Morning Dose: 2

Afternoon Dose: 0

Night Dose: 2

Duration: 7 days

Allergies: no

SHS CLINIC

DIGITAL PRESCRIPTION

Doctor Name : Dr. Ahmed Khan

Patient Checking Day : monday

Patient Checking Date : 04-03-26

Patient Checking Time : 2PM

Patient Name : saim ayyub

Patient Age : 18

Patient Height : 1.83 m

Patient Weight : 65 kg

Patient Fever : 101 celcius

Patient Condition : high fever

Patient Symptoms : cough and throat infection

Patient BMI : 19.41

Health Status : Normal Weight

Medicine Name : burifen tab and panadol

Morning Dose : 2

Afternoon Dose : 0

Night Dose : 2

Duration : 7 days

Allergies : no

Patient 2

|

Conclusion

In conclusion, the Clinic Prescription System is a simple yet powerful project that demonstrates the use of Object-Oriented Programming in solving real world problems. It allows efficient handling of patient data and provides a structured way to generate prescriptions.

This project helped in understanding important OOP concepts like encapsulation, abstraction, and modularity. It also showed how programming can be used to improve real life systems.

